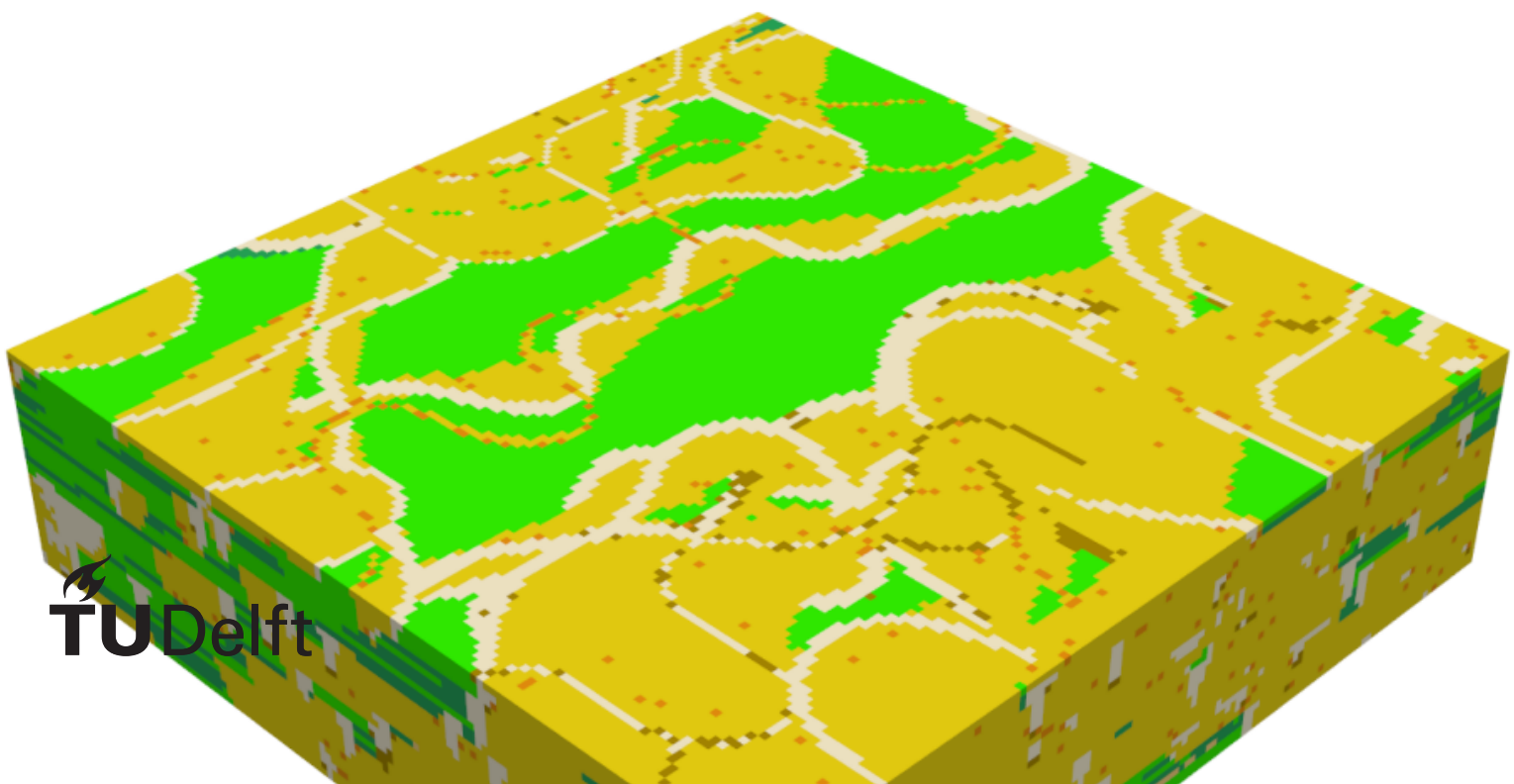


Evaluating the applicability of Generative Adversarial networks to generate geological facies models

A Case study for the Delft Area

M.D. Torres Ruhe



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by

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Abstract

Here is the text of my abstract...

Acknowledgements

I would like to thank...

List of Abbreviations

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cGAN Conditional Generative Adversarial Network. 11

DSST Delft Sandstone. 3, 4

FluvGan Fluvial Generative Adversarial Network. 11

GAN Generative Adversarial Network. 11

TUD Technische Universiteit Delft. 3

TvD True Vertical Depth. 3

WNB West Netherlands Basin. 3

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1 Introduction

Sedimentary deposits represent a vital commodity in modern society due to the various resources that can be extracted from them. These resources include minerals, groundwater, geothermal heat, sands and hydrocarbons (e.g., Brekke et al., 2001; Donselaar et al., 2017; Donselaar et al., 2015; Morris & Ramanaidou, 2007). Moreover, recent studies have investigated storage possibilities for carbon dioxide and hydrogen in this subsurface environment (Heinemann et al., 2021; Zhou et al., 2020). Beyond the increase in subsurface applications, the world also deals with an increasing population and energy demand. Consequently, the demand for resources hosted within fluvial sedimentary deposits is expected to increase while the distribution of the commodities themselves becomes sparser and of lower quality (e.g., Calvo et al., 2016; Spittler et al., 2020). This highlights the need for a deeper understanding of subsurface environments to enable more precise resource management which can be achieved through various methods such as: data gathering surveys, geological evolution and constructing high-quality 3D geomodels.

3D geomodels are traditionally constructed by engineers and geoscientists to predict reservoir properties and resource distribution (Royer et al., 2015; SONG et al., 2022). The voxels in these models may be populated by continuous or categorical properties: grain size, porosity & permeability values or facies associations. These predictions are fundamentally based on a foundational geological conceptual model. The conceptual model acts as a hypothesis regarding the most likely rock mass configuration, built entirely from sparse field data (Koltermann & Gorelick, 1996). This data sparsity is an unavoidable reality of subsurface exploration, resulting from the exceptionally high costs and extended runtimes required for physical data acquisition. Consequently, engineers develop a working understanding of the subsurface based on limited information. This reliance on limited information drives the continuous need to improve geomodelling techniques. Such improvements aim to computationally bridge the gap between isolated data points and realistic, continuous geological representations.

1.1. Problem definition

Despite ongoing advancements in data collection, data scarcity dictates that functional 3D geomodels must rely heavily on subjective interpretations of the underlying geological conceptual model. Because transitioning from these interpretive hypotheses into a digital framework requires applying specific mathematical and spatial algorithms, this workflow inevitably introduces uncertainty at multiple stages, stemming from both the sparse data inputs and the modelling steps themselves (Høyer et al., 2024). As a result, precisely measuring its extent is crucial for effective decision-making and can be accomplished with various techniques (e.g. Monte Carlo Simulations), necessitating the creation of a large collection of equally likely 3D geomodels. Creating this extensive ensemble, however, reveals a core drawback in conventional modeling techniques: the significant compromise between computational efficiency and geological precision. To sustain the generation rate needed for extensive uncertainty quantification, traditional methods frequently depend on greatly simplified, weak geological assumptions.

Due to the fact that these approaches are often limited to simple statistical distributions or strict object-oriented rules (Rongier & Peeters, 2025b), they find it challenging to replicate the intricate, non-linear architectural relationships present in natural sedimentary formations (Sun et al., 2023). As a result, existing workflows compel engineers to decide between models that can quickly quantify uncertainty but do not possess geological realism, or highly precise multi-conceptual models that require too much effort to produce in large quantities (Enemark et al., 2019). This bottleneck highlights a significant deficiency in contemporary subsurface characterization. A significant need exists for a modeling approach that can thoroughly incorporate complex geological prior knowledge while also ensuring the fast generation rates necessary for ensemble-based uncertainty quantification. As a result, the inquiry comes up regarding how the existing modelling process can be modified to ensure that:

- 1) Previous geological insights and structural intricacies are better employed.

- 2) Uncertainty in the produced models is quantified more precisely and efficiently.

To address the need for both geological realism and rapid ensemble generation, recent research has proposed advanced machine learning algorithms as a highly effective solution. These algorithms, specifically the modern subset known as Deep Learning, possess the unique ability to learn complex, non-linear spatial relationships directly from data. By learning these intricate relationships, Deep Learning can accurately capture the geological complexity and non-stationarity that traditional methods often miss (Rongier & Peeters, 2025b). Furthermore, once such a model is fully trained and validated, the generative subset of Deep Learning models can generate a vast ensemble of 3D realizations at a fraction of the time compared to slower process-based models. This rapid generation directly facilitates the large-scale uncertainty quantification demanded by modern subsurface engineering.

While the broader application of machine learning in geomodelling has historical roots (Demyanov et al., 2008), recent surges in computing power have yielded highly capable 3D **Deep Generative Modelling (DGM)** architectures, including **Variational Auto Encoders (VAEs)** and **Latent Diffusion Models** (e.g., Bhavsar et al., 2024; Di Federico & Durlofsky, 2025; Laloy et al., 2017; Song et al., 2023). Among these varied architectures, this study specifically focuses on **Generative Adversarial Networks**. While recent literature highlights **Latent Diffusion Models** as a vanguard for precise hard data conditioning, they inherently suffer from slow inference speeds due to their iterative denoising process. In contrast, **Generative Adversarial Networks** offer near-instantaneous sampling speeds perfectly suited for large-scale Monte Carlo uncertainty loops, despite well-known training vulnerabilities such as mode collapse. **Generative Adversarial Networks** are therefore selected due to their computational efficiency, established research history in spatial generation, and the wide availability of architectures strictly tailored for geomodelling. However, successfully training such **Generative Adversarial Networks (GANs)** requires an extensive volume of high-quality data to act as the geological prior. To overcome the inherent scarcity of real subsurface data, process-based forward modelling software, such as **FLUMY** (Martin et al., 2016), can be utilized to generate robust synthetic training datasets.

1.2. Research gap and opportunity

Utilizing these synthetic datasets allows **Generative Adversarial Networks** to effectively learn geological priors; however, a critical research gap remains regarding their ultimate deployment. While **DGM** excels in theoretical environments, recent literature emphasizes an urgent need to transition from idealized, synthetic test cases to practical field applications. As Rongier and Peeters (2025b) stated, “We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs”. Similarly, Di Federico and Durlofsky (2025) noted that the application (and extension as required) of the overall methodology to real cases should be performed. Addressing this explicit mandate forms the core opportunity of this research. Consequently, the primary goal of this project is to bridge the divide between synthetic training environments and practical field operations by assessing the performance of deep generative modelling in a more complex, real-world application.

Add: section with the full opportunity for this study

Testing these models in a real-world application requires targeting an active subsurface system, a role fulfilled in this study by utilizing data from the Delft Sandstone reservoir. This reservoir target must be characterized using a simple geological conceptual model combined with sparse well data. Such sparse well data will serve as the hard conditioning constraints to evaluate how the generative model behaves under tight physical limitations. These limitations, and the resulting fluctuations in uncertainty, will be analyzed by testing the workflow across multiple distinct geological priors. Investigating the interplay between varied priors and sparse conditioning constraints ultimately establishes the framework for the specific objectives of this study.

1.3. Objective and research questions

With the research gaps in mind, the main goal of this thesis will be to assess the applicability of using a Generative Adversarial Network to generate realizations of a real-world reservoir by conditioning it to well data. Based on the outcome of this thesis, new insight can be given into the workflow needed to approach applying and evaluating Generative Adversarial Networks for generating subsurface reservoir realizations. Hence, the main research question of this work is: *To what extent can Generative Adversarial Networks be used to create geologically plausible realizations of the Delft Sandstone reservoir while conditioned to sparse real-world data?* To help answer this question, several sub-research questions have been formulated:

1. How should the Delft Sandstone's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modeling?
2. Which combination of hyperparameters yields a stable training process?
3. To what degree is the model able to reproduce the complexity inherent in the training data?
4. To what degree does the model generate realizations that are successfully conditioned to well data?
5. How different do models trained on different geological priors react when sparse data is introduced?

1.4. Thesis outline

The remainder of this thesis is structured to systematically address these research objectives. Chapter 2 establishes the theoretical foundation by reviewing the principles of geological modelling, the mechanics of deep generative models, and the specific geological setting of the Delft Sandstone Member. Building upon this theoretical background, Chapter 3 outlines the core workflow, describing the generation of synthetic training datasets using FLUMY and detailing the specific Generative Adversarial Network (GAN) architecture deployed for the reservoir modelling tasks. The empirical outcomes of this workflow are presented in Chapter 4, which evaluates the training process and analyzes the structural and spatial realism of the generated realizations. These results are critically examined in Chapter 5, which contextualizes the findings, addresses the inherent limitations of the proposed approach, and highlights avenues for future research. Finally, Chapter 6 summarizes the main contributions of this thesis and provides definitive answers to the central research questions.

2 Background

This chapter establishes the theoretical and geological foundations and reasoning upon which this research is built. It provides a short overview of the geological setting of the Delft reservoir and a literature review of modern geomodelling and machine learning (ML) techniques. Section 2.1 introduces the geological context of the Delft reservoir, detailing its depositional environment and current conceptual model. Section 2.2 explores the state-of-the-art in geological modelling, comparing traditional techniques and their respective limitations. Finally, Section 2.3 provides a general overview of machine learning and deep learning (DL) principles, focusing on their historical evolution and current application to subsurface characterization. This balanced overview is intended to provide both geological and computer science readers with the necessary context to understand the research methodology and results presented in subsequent chapters.

2.1. The Delft Geothermal Reservoir

The data utilized in this study is sourced from the Delft geothermal reservoir, situated beneath the [Technische Universiteit Delft \(TUD\)](#) Campus in the Netherlands. As of February 2026, a geothermal doublet is actively producing heat from this reservoir (Beerda, 2026). A doublet system consists of a two-well setup: a producer well extracts warm formation water, and an injector well returns the cooled water to the reservoir. At the Delft site, the producer well (**DEL-GT-01**) was drilled to a [True Vertical Depth \(TvD\)](#) of approximately 2,337 m, while the injector well (**DEL-GT-02-S2**) reached a TvD of approximately 2,100 m (Vardon et al., 2024). The project serves both commercial and research objectives, acquiring extensive geophysical logs and core data, supplemented by surface and downhole monitoring instrumentation (Vardon et al., 2020; Vardon et al., 2024). Despite the high data density, the cored section is restricted to the first 80 m of the **DEL-GT-01** well (approximately one-fourth of the reservoir thickness), while **DEL-GT-02-S2** contains no cored sections. The primary target is the [Delft Sandstone \(DSST\)](#), a member of the Lower Cretaceous Nieuwerkerk Formation in the [West Netherlands Basin \(WNB\)](#) (Donselaar et al., 2015).

2.1.1. Geology of the Delft Reservoir

The [WNB](#) is a 60 km wide transtensional basin in the southwest of the Netherlands, characterized by NW-SE trending structural highs and depressions (Donselaar et al., 2015; Ziegler, 1990). The [WNB](#) entered an active rifting stage during the Middle Jurassic (Bodenhausen & Ott, 1981; Den Hartog Jager, 1996; Willems et al., 2020), leading to the formation of several half-grabens filled with terrestrial sediments sourced from neighboring regions (Den Hartog Jager, 1996). The progressive filling of these half-grabens coincided with a change in the depositional environment as a transgressive shoreline moved the system from fluvial to marine. The Nieuwerkerk Formation comprises three members: the Alblasserdam, the [DSST](#), and the Rodenrijs Claystone.

The Alblasserdam member was deposited in a fluvial environment characterized by stacked channels and floodplain deposits (DeVault & Jeremiah, 2002). The [DSST](#) directly overlays the Alblasserdam and consists of thick, stacked fluvial-deltaic channel sandstones with high net-to-gross values, oriented NW-SE (Bruining, 2023; Groenenberg et al., 2010; Mijnlieff, 2020; Willems et al., 2020). The [DSST](#) is in turn overlain by the Rodenrijs Claystone, an organic-rich lagoonal deposit (Bruining, 2023; DeVault & Jeremiah, 2002; Willems et al., 2020).

Based on the sedimentological analysis of the **DEL-GT-01** core and **DEL-GT-02-S2** sidewall samples, three primary facies associations (FAs) have been identified (Bruining, 2023):

- **FA1: Fluvial and distributary channel deposits.** Primarily light grey, fine- to medium-grained sandstones featuring low-angle cross-bedding and ripple lamination, with unit thicknesses ranging from 5 to 80 m.

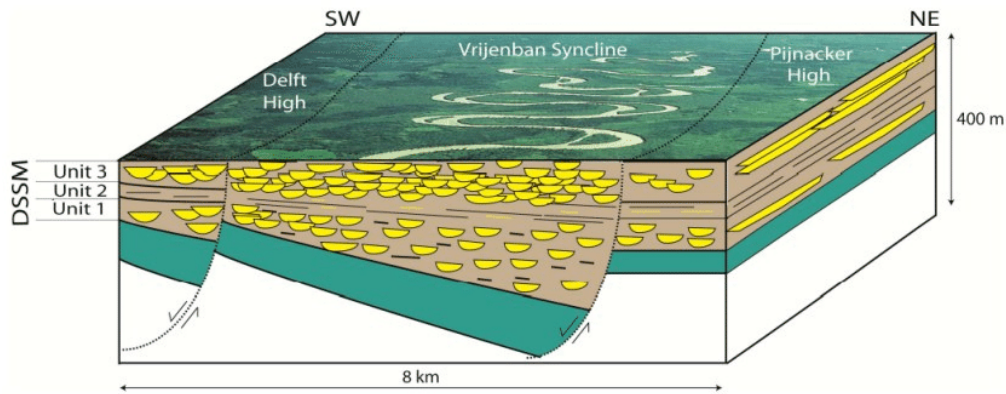


Figure 2.1: Depositional models for the DSST Member with three identified units. Accommodation space is increasing towards the NE (Donselaar et al., 2015)

- **FA2: Crevasse splays and levee deposits.** Composed of moderately sorted, very fine- to medium-grained sandstones interbedded with dark grey siltstones, with thicknesses ranging from 1 to 5 m.
- **FA3: Overbank deposits and paleosols.** Consists of dark grey siltstones, claystones, and very fine-grained sand deposited in low-energy environments. This association is characterized by abundant organic material, slickensides, and siderite nodules.

The specific depositional environment of the DSST remains a subject of scientific discussion. It has been attributed to both fluvial-meandering (Donselaar et al., 2015; Vondrak et al., 2018; Willems et al., 2020) and fluvio-deltaic systems (Bruining, 2023; van Adrichem Boogaert & Kouwe, 1993; Weert et al., 2024). Recent analysis by Bruining (2023) suggests that the cored section represents a deltaic environment. The observed facies associations can be attributed to a upper to lower plain delta with multi-stacked fluvial/distributary channels, single channels, crevasse splays, levees and overbank fines.

2.1.2. Deltaic Depositional Environment

Deltaic environments form at the land-sea interface where river systems supply clastic sediments. Delta morphology is governed by the relative influence of fluvial, wave, and tidal processes (Kolb & Van Lopik, 1966). The subaerial delta plain is typically divided into two domains (Heldreich et al., 2017; Kolb & Van Lopik, 1966):

- **Upper Delta Plain:** Lies above the limit of saltwater intrusion. Sedimentation is driven by distributary channel migration, overbank flooding, and crevasse splaying. Environments include meandering and braided channels, lacustrine fills, and floodplains.
- **Lower Delta Plain:** Subject to river-marine interaction and tidal influence. Channels are more numerous and exhibit bifurcating or anastomosing patterns. Interdistributary bays, bay fills (crevasse splays), and marshes dominate the landscape.

In the DSST, FA1 features organic material and siltstones indicating slackwater periods, while the lack of bioturbation suggests high sedimentation rates. The fine-grained nature of FA2 and FA3 reflects more distal positions on the delta plain. Siderite nodules in FA3 indicate anoxic, waterlogged conditions (Samuel, 1959), while soft-sediment deformation suggests rapid loading of mudstones over saturated sands. These features collectively indicate that FA3 represents deposition within a delta plain, likely in an interdistributary bay or a pro-delta environment.

2.2. Geological Modelling

The primary objective of 3D reservoir modelling is to provide a representative volume of the subsurface for flow simulation and decision support. These models are dynamic, requiring updates as new data becomes available to refine predictions over the reservoir's lifetime (Emerick, 2025; Ringrose & Bentley, 2016). Existing techniques are generally categorized into three types (Koltermann & Gorelick, 1996):

- **Pixel-based Models:** Use spatial statistics (e.g., variograms) for cell-by-cell property assign-

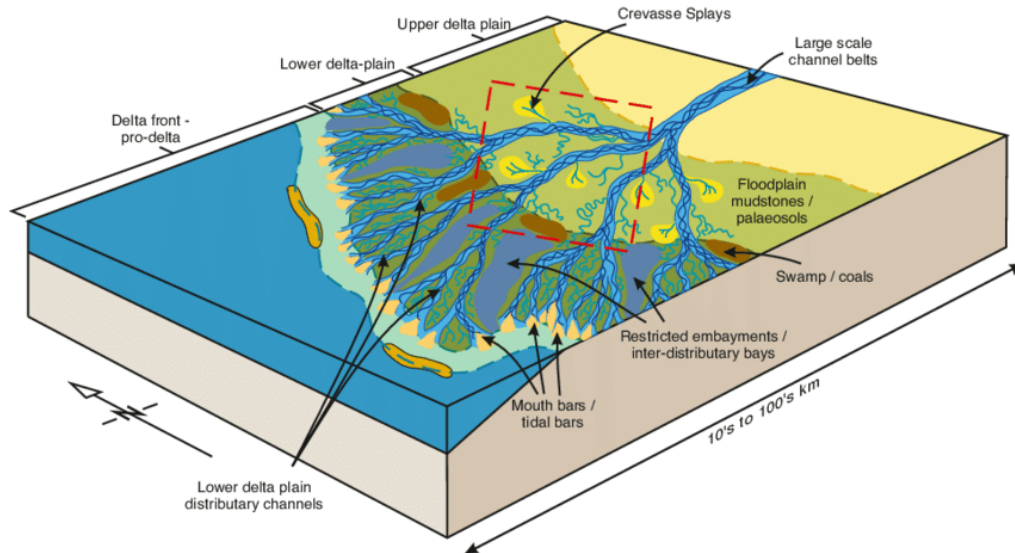


Figure 2.2: Depositional model of the Mungaroo Formation, illustrating the geometries and domains of a fluvial-deltaic system (Heldreich et al., 2017).

ment. They are efficient and honor dense well data but often fail to capture complex connectivity, resulting in pixelated realizations.

- **Object-based Models:** Directly place geometric features (e.g., channels) based on geological rules. They offer high visual realism but struggle to honor dense or closely spaced well data.
- **Process-based Models:** Simulate the processes behind sediment transport and deposition. They provide the highest degree of realism but are computationally expensive and difficult to condition to specific hard data like well logs.

Attribute	Geostatistical	Object-Based	Process-Based
Geological Prior	Variograms, training images.	Geometric shapes and rules.	Physical boundary conditions.
Realism	Low to Moderate.	Moderate to High.	Very High.
Key Advantages	Computational speed, well conditioning.	Realistic connectivity.	Physical consistency.
Key Disadvantages	Lack of geological realistic shapes.	Difficult conditioning.	High computational cost.

Table 2.1: Comparison of spatial modelling techniques in geomodelling.

While traditional pixel-based geomodelling relies heavily on two-point spatial statistics, such as variograms, to distribute properties, these methods often fail to capture the complex, multipoint connectivity inherent in natural geological systems. Machine learning introduces a paradigm shift by moving away from user-defined spatial rules. Instead of requiring an engineer to pre-define the shape of a channel or the range of a variogram, ML algorithms learn these intricate, non-linear spatial relationships directly from data. This makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are too complex to be captured by standard statistical interpolation.

2.3. Machine Learning

Machine learning (ML) involves the development of algorithms that learn patterns from data to perform specific tasks without explicit programming (Goodfellow et al., 2016). This foundational concept

of learning without explicit programming has evolved significantly since Samuel (1959) introduced the term, expanding into a wide array of computational techniques. These diverse techniques are fundamentally categorized based on their training approach into two primary domains:

- **Supervised Learning:** The algorithm is trained on a labeled dataset containing explicit input-output pairs.
- **Unsupervised Learning:** The algorithm identifies hidden patterns within unlabeled data without prior knowledge of the desired output.

Within these two broad categories, algorithms are deployed to solve common tasks such as classification, regression, clustering, and generation. Applying these generative tasks to the subsurface highlights a severe limitation in traditional pixel-based geomodelling, which relies heavily on simple two-point spatial statistics (e.g., variograms) to distribute properties. Because these simple statistics often fail to capture the complex, multipoint connectivity inherent in natural geological systems, ML introduces a necessary paradigm shift. This shift moves the modelling workflow away from strict, user-defined spatial rules. Instead of requiring an engineer to pre-define the range of a variogram, ML algorithms learn intricate, non-linear spatial relationships directly from the data. Learning directly from data makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are simply too complex to be captured by standard statistical interpolation. The capacity to uncover such complex, non-linear relationships in vast datasets is largely driven by modern [Artificial Neural Networks \(ANNs\)](#) (LeCun et al., 2015). Inspired by the biological structure of the brain, these [ANNs](#) consist of interconnected artificial neurons that process information across multiple computational layers (Goodfellow et al., 2016).

While [ANNs](#) and traditional machine learning algorithms excel at analyzing tabular petrophysical data for various subsurface tasks, they inherently lack actual geological reasoning skills. Without such reasoning to naturally make sense of areas with low data quality, these algorithms are restricted to inferring abstract features directly from their training data ([schaaf2019quantification](#)). Inferring these features in standard ML workflows necessitates manual feature extraction, a process where engineers must explicitly define the relevant geological parameters prior to training. This strict reliance on manual feature extraction creates a critical limitation when attempting to scale these algorithms to 3D geomodels. Within these high-resolution 3D geomodels, which typically contain millions of voxels, manually defining multi-scale spatial features ultimately becomes computationally impractical.

Deep Learning (DL) overcomes this limitation through automatic feature extraction. By utilizing multi-layered networks, DL architectures natively process high-dimensional spatial grids, bypassing the manual extraction bottleneck. For subsurface characterization, this means a DL model can ingest raw 3D conceptual models or process-based simulations and autonomously identify the underlying geological rules. This capability to scale with high-dimensional data makes DL the premier mathematical framework for modern reservoir modelling.

2.3.1. Deep Learning and Inductive Biases

Deep learning (DL) is a subset of ML focusing on multi-layered neural networks that represent data with multiple levels of abstraction (Goodfellow et al., 2016; LeCun et al., 2015). Achieving these levels of abstraction in high-dimensional spaces introduces a fundamental challenge known as the “curse of dimensionality.” This curse of dimensionality describes the exponential increase in data required as the spatial dimensions grow, making high-dimensional estimation computationally challenging. To overcome this computational challenge and help models generalize from limited data, engineers implement specific architectural choices known as inductive biases (Prince, 2023). A premier example of such an inductive bias is the foundational architecture used for processing spatially structured data: the [Convolutional Neural Network \(CNN\)](#).

[CNNs](#) leverage spatial inductive bias to become highly effective for 2D and 3D geomodelling tasks. Executing these spatial tasks relies on a core architectural innovation known as the convolutional layer. This convolutional layer replaces the fully connected neurons of a standard [ANN](#) with a localized set of learnable filters, commonly referred to as kernels. These multi-dimensional kernels operate by systematically sliding across the input data in discrete increments dictated by a predefined step size, or stride. This sliding stride is frequently combined with padding, a technique that adds artificial zero-

value borders to the input to ensure the entire spatial region is comprehensively analyzed. Analyzing this spatial region involves computing a local dot product between the kernel weights and the input values, a mathematical operation visually illustrated in Figure 2.3.

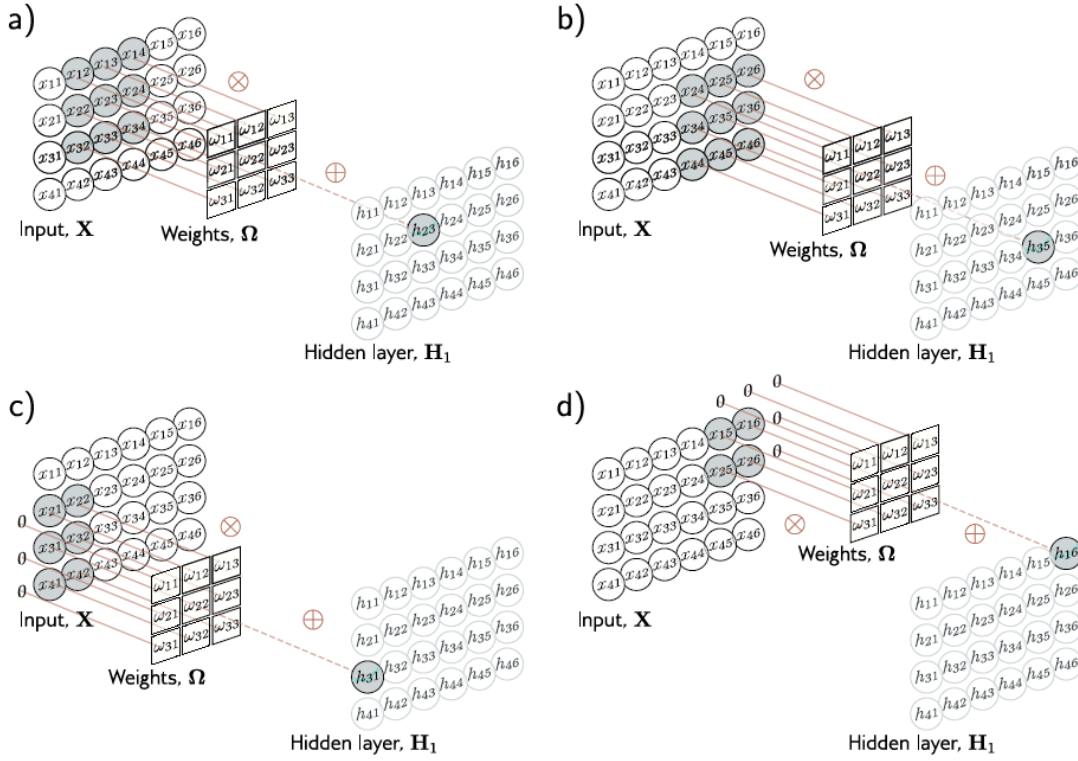


Figure 2.3: Example of a 2D convolutional layer with kernel size of 3x3 and a padding of 1. Each output h_{ij} is composed of a weighted sum of the 3x3 nearest inputs on the previous layer (Prince, 2023). a) Here, the output h_{23} (shaded output) is a weighted sum of the nine positions from x_{12} to x_{34} (shaded inputs). b) Different outputs are computed by translating the kernel across the image grid in two dimensions. c–d) With zero-padding, positions beyond the image's edge are considered to be zero.

The mathematical result of this operation is a new, filtered spatial representation known as a feature map. This feature map is typically downsampled by a subsequent pooling layer, which condenses the spatial dimensions while preserving the most prominent information. Preserving this information across a continuous stack of convolutional and pooling layers allows the network to progressively map low-dimensional input data into high-dimensional, abstract features. At the base of this feature mapping process, the initial layers capture simple, low-level spatial patterns, such as discrete edges or sharp facies boundaries. By mathematically combining these simple low-level edges in deeper layers, the network learns to recognize highly complex, high-level geological architectures, such as the overall sinuosity of a fluvial channel. Learning these complex high-level architectures is computationally efficient because the exact same kernel is applied universally across the entire grid, creating an inherent advantage known as translation invariance. This translation invariance ensures that if the network learns to identify a channel structure in one corner of the reservoir, it can accurately recognize that same structure anywhere else in the model, drastically reducing computational overhead and preventing overfitting. Further preventing this overfitting requires another critical component of the DL workflow: the strict use of validation and testing protocols. These validation protocols ensure that the hierarchical patterns learned by the network are robust and safely applicable to unseen data. Applying these robust, learned spatial patterns to unseen subsurface data forms the fundamental basis of modern **Deep Learning (DL)**-driven geomodelling workflows.

2.3.2. Geomodelling with Deep Learning

Modern DL-driven geomodelling workflows have evolved significantly, transitioning from simple pixel-based interpolations to the generation of highly sophisticated spatial representations. Generating these sophisticated representations relies on the ability of DL to directly learn complex geological features from large datasets or process-based simulations. Learning directly from such simulations bridges the historical gap between simplistic descriptive statistics and true geological realism (Laloy et al., 2017). To achieve this realism, CNNs have proven particularly successful in capturing the complex, non-stationary architectures of fluvial channels and facies distributions (Sun et al., 2023). Capturing these intricate architectures enables the rapid generation of stochastic realizations that successfully honor both the global geological concept and local, sparse conditioning data. Honoring these dual constraints while generating massive ensembles of stochastic realizations fundamentally enhances subsurface uncertainty characterization (Song et al., 2023). This unique capacity for rapid, constrained generation is primarily driven by specialized neural network frameworks, most notably generative architectures.

These generative architectures that are using CNNs, are best known for their ability to generate realistic images. The most-well known models include VAEs, GANs and Latent Diffusion Models. Though they are all different in terms of architecture, some core concepts remain similar. Generative models are designed to learn and approximate the complex joint probability distribution of the training data whereby elements are generated based on previously generated values (he2025generative). Moreover, these models use a latent space, designed to either store the encoded training data (VAEs) or be used to scale to an image through various upsampling steps (GANs and Latent Diffusion Models) (Prince, 2023).

2.3.3. Generative Adversarial Networks

Generative architectures, specifically Generative Adversarial Networks (GANs), achieve complex spatial generation through a competitive framework between two distinct neural networks (Goodfellow et al., 2014). These two networks are defined as a Generator (**G**) and a Discriminator (**D**). The generator is tasked with creating synthetic spatial samples from random noise, while the discriminator is simultaneously tasked with distinguishing these synthetic samples from actual real-world training data. The continuous competition to outsmart one another drives a minimax game, which is mathematically expressed as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{D}, \mathbf{G}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log \mathbf{D}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - \mathbf{D}(\mathbf{G}(\mathbf{z})))] \quad (2.1)$$

In this equation, $\mathbf{x}$ represents the real data, and $\mathbf{z}$ represents a random latent vector sampled from a prior distribution $p_{\mathbf{z}}$. The first term represents the discriminator successfully recognizing real geological data, while the second term represents the discriminator successfully catching fake data generated by $\mathbf{G}(\mathbf{z})$. During training, the discriminator seeks to maximize this overall value function to perfect its detection abilities, while the generator seeks to minimize it to successfully fool the discriminator. By successfully fooling the discriminator, the generator ultimately learns to produce high-fidelity geological realizations without requiring any explicit, hard-coded spatial rules. An overview of the training process is provided in figure B.2.

GAN Inversion

Figure B.2 illustrates the generation of high-fidelity, unconstrained realizations, which is a powerful capability; however, practical reservoir modelling requires these models to honor sparse physical constraints like hard well data. Hard well data constraints are typically honored through an optimization technique known as GAN inversion. GAN inversion operates strictly within the model's latent space after the training phase is complete. Completing the training phase allows the algorithm to iteratively search and optimize this latent space to identify a specific latent vector $\mathbf{z}$. This specific latent vector $\mathbf{z}$ generates a 3D volume perfectly matching the sparse well data. Matching the sparse well data securely conditions the generative model to real-world subsurface observations.

Wasserstein Metric for Loss and Validation

Real-world subsurface observations used to condition the realizations must be rigorously evaluated to ensure geological realism and consistency. This consistency in high-dimensional 3D spatial data is historically computationally difficult to evaluate due to the curse of dimensionality. The curse of

dimensionality can be mitigated by utilizing the Wasserstein distance as a robust validation metric and training loss function. This training loss function calculates the optimal transport cost, or Earth Mover's Distance, between the true data probability distribution and the generated probability distribution. The generated probability distribution's deviation from the true distribution is mathematically defined by the basic Wasserstein formula:

$$\mathcal{W}(P_r, P_g) = \inf_{\gamma \sim \Pi(P_r, P_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|] \quad (2.2)$$

This basic formula calculates the minimum cost required to transform the generated distribution P_g into the real distribution P_r . The real distribution P_r is approximated efficiently through this calculation, allowing the network to converge on highly realistic, non-linear structures even when exact pixel overlap is minimal.

Mean Sliced Wasserstein Distance

Minimal exact pixel overlap across massive 3D grid volumes makes direct mathematical comparison highly inefficient, necessitating refined validation metrics such as the Mean Sliced Wasserstein Distance (MSWD). MSWD bypasses this high dimensionality by mathematically projecting the 3D facies distributions onto multiple 1D slices. These 1D slices provide a highly efficient framework for distance comparison. This distance comparison ultimately validates whether the newly generated geomodels remain geologically consistent with the original training data.

3 Methodology

This chapter describes the comprehensive project workflow, beginning with the generation of 3D synthetic datasets used for training, validating, and testing the models (Section 3.1). Section 3.2 details the GAN framework, including the network architectures, the facies representation strategy, and the loss functions employed during training. Finally, Section 3.3 introduces the metrics and simulation methods used to evaluate the geological realism and practical utility of the generated realizations.

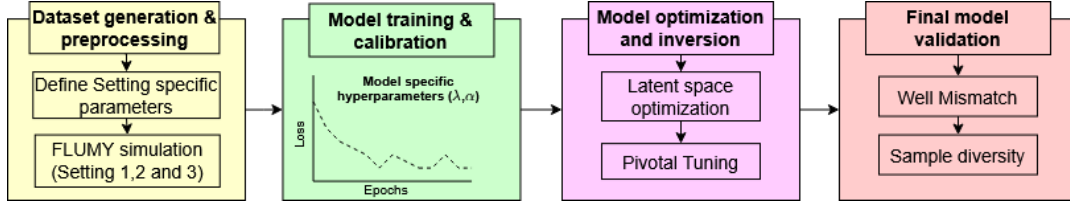


Figure 3.1: General overview of the workflow implemented in this project. (enhance)

3.1. Dataset Generation

In order to rigorously test the applicability of GANs several new datasets had to be generated. Training deep learning algorithms for 3D geomodelling requires a substantial volume of high-fidelity training data. The new datasets were generated using FLUMY (Martin et al., 2016), a process-based stochastic modelling software designed to reproduce meandering channelized reservoirs. Meandering channelized reservoirs are accurately replicated because FLUMY simulates facies, relative grain size, and depositional time for each voxel in a 3D grid based on specified hydraulic and sedimentological rules. Geologically realistic features are produced without requiring excessive computational overhead, making the software not very computationally intensive thus making it possible to generate a large number of models in a relatively short amount of time.

The FLUMY python distribution is based on Non-Expert User Settings (NEXUS) that only allow three customizable parameters: 1) Maximum Channel Depth, 2) Maximum Sand Body Extension, and 3) Net to Gross. Based on these three customizable inputs, the rest of the parameters as provided in the user guide is estimated automatically. Although the software is specifically designed to simulate fluvial meandering rivers, its underlying process-based parameters can be intentionally adapted to mimic alternative depositional domains. Mimicking these alternative depositional domains allows the software to effectively capture the complex deltaic architectures hypothesized for the Delft Sandstone reservoir.

3.1.1. Dataset Descriptions

The three distinct geological conceptual models developed to capture the range of environments hypothesized for the Delft Sandstone reservoir are compiled into separate training suites. These separate training suites consist of setting 1, 2 and 3. Table 3.1 provides an overview of the NEXUS input parameters for all three settings.

Parameter	Setting 1	Setting 2	Setting 3
Net-to-Gross (%)	67	67	67
Max. Channel Depth (m)	5	6	7
Max. Sand Body Extension	80	100	120

Table 3.1: NEXUS input parameters for the three training datasets.

The hypothesized depositional environments are split across three discrete suites, Ranging from a

Lower plain to Upper plain deltaic characteristics in terms of increasing channel depth and wider channel bodies. Beyond these specific NEXUS constraints, all generated models share a uniform reservoir scale with a total grid resolution fixed at $128 \times 128 \times 32$ voxels. In order to allow for well placement in the samples, the final dimension should encompass an area of at least 2000×2000 m as the **DEL-GT-01** and **DEL-GT-02-S2** have an 1500m spacing between them. Establishing this specific grid resolution combined with a standard voxel dimension of $20 \text{ m} \times 20 \text{ m} \times 1 \text{ m}$ means that the total simulated domain securely covers a volume of $2560 \text{ m} \times 2560 \text{ m} \times 32 \text{ m}$. Every single voxel within this acute 32 m vertical interval is ultimately encoded with 1 of the 14 distinct facies (see A.1) available in FLUMY, of which not all will be present in the dataset samples due to the input parameter settings.

3.1.2. Sample preprocessing

To comprehensively assess the capabilities of **GANs**, the synthetic training datasets are prepared at two distinct categorical resolutions. These two distinct resolutions consist of a high-fidelity 9-facies dataset and an upscaled 3-facies dataset. The high-fidelity 9-facies dataset preserves the detailed depositional environments generated by FLUMY, serving as a baseline to evaluate the model's pure generative capacity and its ability to replicate highly complex spatial distributions. While evaluating this generative capacity is crucial, practical field application requires conditioning the model to actual subsurface observations. These actual subsurface observations, derived from the cored section of the **DEL-GT-01** well, are strictly classified into three primary facies associations (Section 2.1.1). Because the hard well data only resolves to three distinct associations, the original FLUMY depositional environments must be grouped accordingly to create the upscaled 3-facies dataset. This grouping process aggregates the detailed environments into broader categories, as detailed in Table 3.2.

Value	Facies Key	Color	Description
1	Channel & Point Bar (FA1)		Grouped proximal deposits: Channel Lag (1), Point Bar (2), Sand Plug (3), and Crevasse Splay I (4).
2	Levee & Crevasse (FA2)		Grouped intermediate deposits: Crevasse Channel (5), Crevasse Splay II (6), and Levee (7).
3	Overbank Fines (FA3)		Grouped distal deposits: Overbank (8), Mud Plug (9), Hemipelagic Plug (10), Wetland/Paleosol (11), Draping (12), and Pelagic (13).

Table 3.2: Overview of the upscaled 3-facies grouping used for the generative modelling, aggregated from the original 13 FLUMY depositional environments (See Table A.1 for the original facies).

Beyond this grouping process, the categorical nature of both the 3-facies and 9-facies datasets necessitates a final preprocessing step known as one-hot encoding (Goodfellow et al., 2016). One-hot encoding is critical because standard neural networks process continuous numerical gradients and might otherwise incorrectly interpret integer facies labels as having an ordinal, mathematical relationship. This ordinal misinterpretation is prevented by transforming the original 3D volume into an expanded 4D tensor, where each facies class is assigned its own dedicated spatial channel. Each spatial channel indicates the presence or absence of a specific facies using values of 1 and -1, which has the result that the facies representation of each facies looks like $[-1, 1, -1]$ given that FA2 was present in that specific voxel.

3.2. Fluvgan Framework

3.2.1. Model Architecture and training configuration

The **Fluvial Generative Adversarial Network (FluvGAN)** model, introduced by Rongier and Peeters (2025a), relies on an optimized DCGAN architecture for 3D fluvial facies modelling. This architecture adapts the BigGAN framework (Brock et al., 2019) to manage the complex connectivity and high resolution necessary for realistic reservoir models. These models originally generated continuous values, which differs from this project's need for categorical facies data. Categorical facies data is therefore processed us-

ing the previously described one-hot encoding strategy, which assigns each facies to a separate output channel. These separate output channels dictate the final depth of the generator’s tensor, represented by the variable C , where $C = 3$ or $C = 9$ depending on the selected experimental track.

This dynamic channel depth is accommodated by an architecture comprising a Generator (**G**) and a Discriminator (**D**) built with 3D Residual blocks. These blocks enable the generator to transform a, in this case, 100-dimensional latent vector $\mathbf{z}$ into a $C \times 32 \times 128 \times 128$ volume. This volume begins as a $512 \times 4 \times 4 \times 4$ feature map, created via a 3D transposed convolution, and is then expanded by five residual upscaling blocks. These upscaling blocks accommodate grid anisotropy by doubling spatial dimensions up to 128, while limiting depth doubling to the first three blocks to reach 32. This final spatial resolution is processed through a Softmax activation, ensuring the C facies channels sum to exactly one per voxel. Every generated voxel is then evaluated by the discriminator, acting as a binary classifier to reduce the $C \times 32 \times 128 \times 128$ input to a single scalar logit. This logit is produced through five residual downscaling blocks that iteratively reduce spatial resolution while increasing the channel count to 512. This 512-channel discriminator, alongside the generator, achieves training stability by utilizing spectral normalization and LeakyReLU activations. Training stability is essential for managing the model’s computational scale, which involves roughly 11.6 million trainable parameters for the generator and 2.0 million for the discriminator.

Hyperparameters

Training GANs is notoriously hard, as described in section ..., since it requires both the Generator (**G**) and Discriminator (**D**) to learn with each iteration. As a result, the hyperparameters of the architecture need to be carefully chosen so both components train properly. By testing a long list of possible combinations, the hyperparameter combination achieving the most geologically plausible models was found with the following values:

Hyperparameter	Value	Description
Batch Size	64	Samples per iteration
Kernel Size	$3 \times 3 \times 3$	3D Convolutional filters
Optimizer	Adam	$\beta_1=0, \beta_2=0.999$
Learning Rate (G,D)	$10^{-4}, 4 \times 10^{-4}$	Two-timescale update rule
num iter D	1	Number of consecutive iterations of the discriminator before an iteration of the generator

Table 3.3: Key hyperparameters for the Fluvgan architecture.

Loss Functions

Evaluating the generated realizations begins by analyzing the foundational mathematical metrics, specifically Binary Cross-Entropy (BCE) and Wasserstein Loss. BCE is deployed because the generator processes one-hot encoded facies, effectively turning the spatial generation into a multi-channel binary classification task. This classification task relies on BCE to penalize pixel-wise deviations between the generated probabilities ($\hat{y}_i$) and the true categorical labels (y_i), defined mathematically across N voxels as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

While this equation effectively measures local, pixel-to-pixel accuracy, it structurally fails to capture the broader spatial connectivity of complex geological architectures. Complex geological architectures and global multi-point statistics are instead evaluated using the Wasserstein Loss, often implemented via the Mean Sliced Wasserstein Distance (MSWD). Wasserstein Loss calculates the optimal transport cost (Earth Mover’s Distance) between the true (P_r) and generated (P_g) probability distributions, allowing the network to converge on highly realistic, non-linear structures even when exact pixel overlap is minimal. This structural distance is expressed as:

$$\mathcal{W}(P_r, P_g) = \inf_{\gamma \sim \Pi(P_r, P_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|]$$

3.2.2. Optimization & Inversion

While the 9-facies models are utilized to evaluate unconstrained generative performance, the 3-facies models are deployed to test practical conditioning against the **DEL-GT-01** well data. To successfully condition this 3-facies generative model to real-world observations, the overall inversion workflow relies on two primary mechanisms:

- **Latent Space Optimization:** An iterative search to find the optimal random noise vector $\mathbf{z}^*$ that best approximates the well data.
- **Pivotal Tuning:** A targeted fine-tuning of the generator's mathematical weights to perfectly lock the generated facies to the hard conditioning data.

This two-step inversion process begins with latent space optimization. Latent space optimization freezes the internal weights of the pre-trained generator and utilizes gradient descent to iteratively search for an optimal latent vector $\mathbf{z}^*$. This optimal vector is found by minimizing a categorical cross-entropy loss between the generated facies at the exact spatial coordinates of the well and the true geological observations. These geological observations anchor the optimization, mathematically defined as:

$$\mathbf{z}^* = \arg \min_{\mathbf{z}} \mathcal{L}(G(\mathbf{z})_{\text{well}}, \mathbf{y}_{\text{well}})$$

This minimization yields an optimized vector $\mathbf{z}^*$ that often provides a close structural approximation, but frequently fails to achieve a perfect 100% match due to the inherent mathematical boundaries of the learned latent space. These latent space boundaries are overcome by transitioning to the second optimization phase, known as Pivotal Tuning. Pivotal Tuning uses $\mathbf{z}^*$ as a static anchor, temporarily unfreezing the generator's internal weights θ to make microscopic adjustments. These weight adjustments forcefully push the localized output to match the exact well data, while a strict regularization term prevents the collapse of the broader geological realism, expressed as:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(G_{\theta}(\mathbf{z}^*)_{\text{well}}, \mathbf{y}_{\text{well}}) + \lambda \|\theta - \theta_0\|$$

3.3. Evaluation Metrics

Validating the outputs generated by the network requires a comprehensive suite of mathematical metrics designed to evaluate local precision, global realism, and overall ensemble diversity:

- **Binary Cross-Entropy (BCE) & Wasserstein Loss:** Evaluates pixel-wise classification accuracy and the preservation of global geological structures.
- **Normalized & Jensen-Shannon Entropy:** Measures pixel-wise spatial uncertainty and total ensemble sample diversity.
- **Macro F1-Score:** Quantifies the localized mismatch at the wellbore while accounting for severe facies imbalance.

Geological Realism

In order to validate the model for geological realism without knowing the ground truth model, it is important to use a set of validation metrics which may be used to help with quantifying the geological realism in the models. To achieve this, validation metrics covering global and local structures are needed. To validate the local geological plausibility in a single sample the connectivity of between facies can be analysed.

Beyond evaluating the structural realism of a single generated model, it is critical to quantify the spatial uncertainty across the entire generated ensemble. This ensemble uncertainty is quantified using entropy metrics, beginning with Normalized Shannon's Entropy (**shannon1948mathematical**). Normalized Shannon's Entropy evaluates the pixel-wise entropy across multiple models, scaling the standard entropy equation by the number of classes (k) to strictly bound the metric:

$$H_{norm}(X) = -\frac{1}{\log_2(k)} \sum_{i=1}^k P(x_i) \log_2 P(x_i)$$

This $[0, 1]$ bounding scale allows to visually map exact regions of spatial uncertainty, highlighting areas where the ensemble highly agrees versus areas of deep geological conflict. This 3D grid of spatial conflict is ultimately aggregated into a single definitive scalar using the Jensen-Shannon (JS) divergence (**lin1991divergence**). JS divergence measures the pixel-wise distributional disagreement and stochastic spread between the probability distributions of the flumy and GAN models, acting as a final quantitative measure of spatial fidelity:

$$D_{JS}(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M)$$

where P and Q represent the categorical probability distributions of the flumy and GAN models respectively, $M = \frac{1}{2}(P+Q)$ and D_{KL} represents the standard Kullback-Leibler divergence (**kullback1951information**).

Data conditioning

While JS divergence defines the overall ensemble spread, each individual GAN model within that ensemble must still strictly honor the physical reality at the wellbore. This physical reality at the wellbore acts as a strict spatial constraint, the localized mismatch of which is calculated using the Macro F1-score. The Macro F1-score calculates the unweighted average of the harmonic mean of precision and recall across all C individual facies classes:

$$F1_{macro} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

Evaluating every facies class independently provides a highly robust performance metric that heavily penalizes the model if it ignores underrepresented minority classes. These underrepresented minority classes, such as the thin crevasse splays (FA2), would otherwise be completely overshadowed by the massive channel sand deposits (FA1) if standard statistical accuracy was deployed to assess the well conditioning.

4 Results

The presentation of the results will follow the order of the workflow as defined in figure 3.1: First of all the curated datasets will be displayed with the addition of statistical parameters that can be used to evaluate the difference between the samples in the datasets and the synthetic GAN samples. Secondly, the post training results are presented which presents the results of the hyperparameter tuning, the cGAN unconstrained performance to grouped facies classes as well as the cGAN unconstrained performance when presented with all facies classes. Finally, the results of the optimization and inversion results are presented, after the model was constrained to the Delft Geothermal Reservoir well data.

4.1. Datasets

In total, 20,000 samples were generated for the three datasets as defined in table 3.1. These generated datasets are visually summarized in figure 4.2, which displays a representative 3D sample block paired alongside a corresponding plot of the normalized spatial entropy. This normalized spatial entropy maps show how the architectural features progress across the training suites, moving from tightly confined, small meandering channels into much deeper channel paths embedded within wider sand bodies. These wider channel architectures heavily influence the accompanying entropy plots on the right, which show that the physical distribution of the facies almost reaches the maximum theoretically possible entropy threshold.

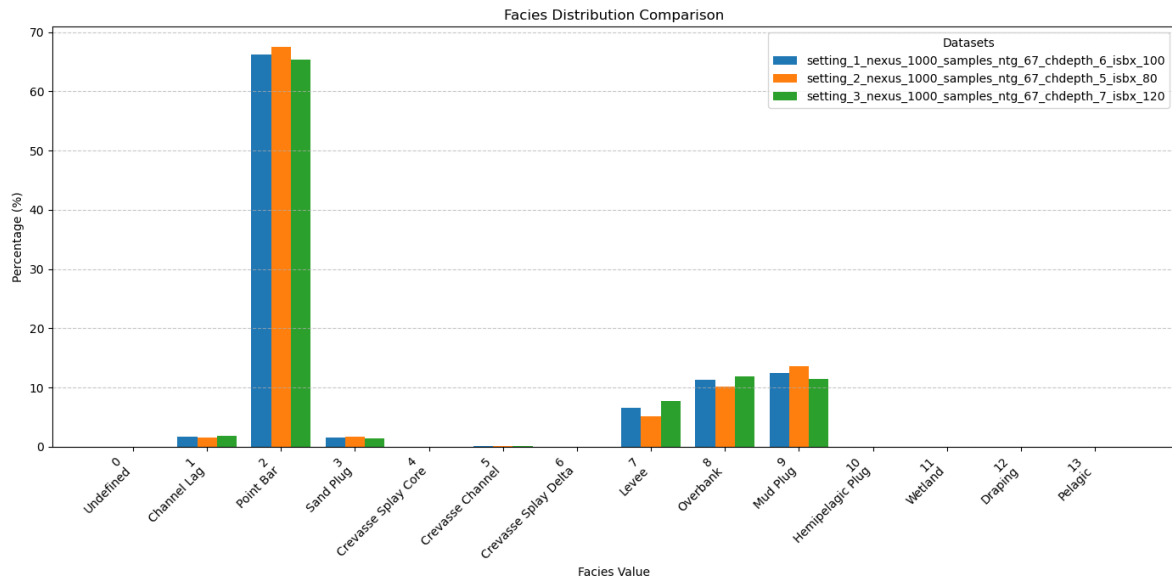


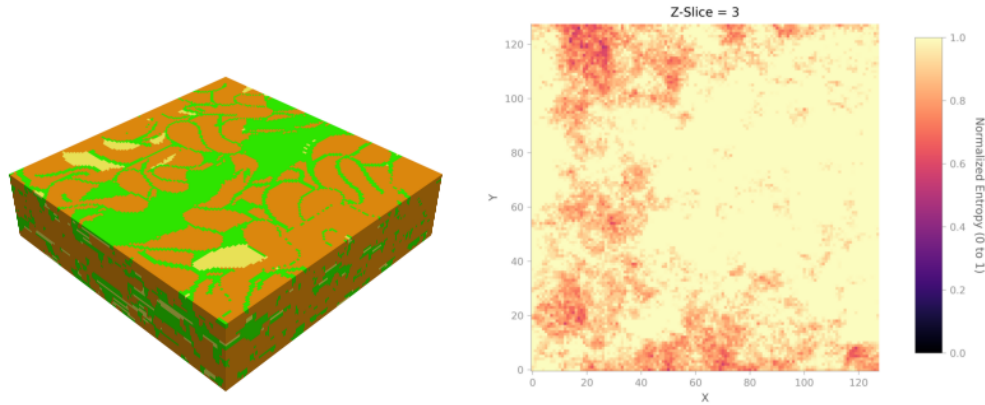
Figure 4.1: Global categorical facies distribution profiles compared across the three synthetic training datasets for all available classes.

This threshold configuration can be further evaluated by analyzing Figure 4.1, which provides an exhaustive overview of the global facies distributions across the ensemble. As can be observed in this overview, the structural facies, hereafter referred to as classes, exhibit a severe statistical imbalance throughout the data volume. This severe imbalance manifests as dominant classes occupying over 60% of the total grid volume, while corresponding minority classes are entirely absent due to the fundamental boundary limitations of the FLUMY NEXUS parameters. The volumetric presence of classes 4, 5, and 6 is particularly suppressed, yielding an exceptionally low occurrence rate ranging between 0.0% and 0.1%. Such depressed occurrence rates present an optimization bottleneck for the subsequent model inversion step, as established sedimentological literature indicates that the real-world occur-

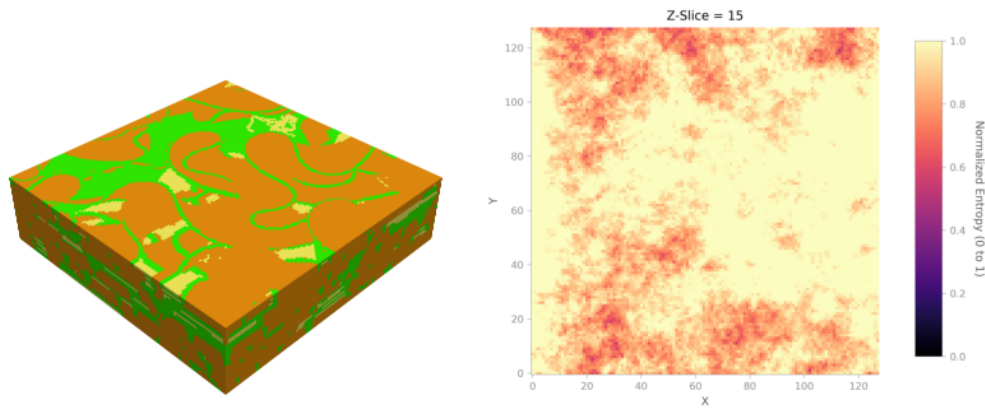
rence of the intermediate Levee & Crevasse association (FA2) within the Delft reservoir is significantly higher, at approximately (...%).

Moreover, the differences of facies distributions between the facies are not substantial. This makes it less intuitive to judge the performance of the model to choose a specific dataset that fits the well data better based off the facies distributions.

a) Setting 1



b) Setting 2



c) Setting 3

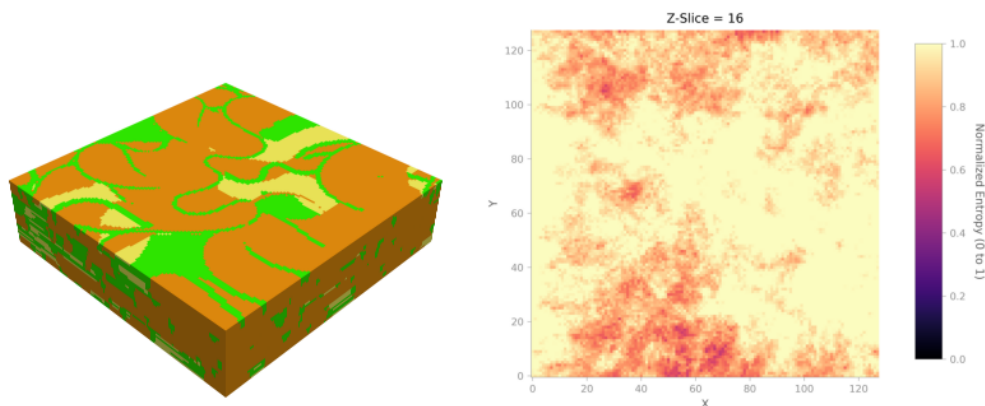


Figure 4.2: Volumetric overview of the synthetic datasets generated with FLUMY, showcasing the structural progression across **a) Setting 1**, **b) Setting 2**, and **c) Setting 3**.

4.2. Post Training Generative Results

4.2.1. Hyperparameter Tuning

4.2.2. cGAN Unconstrained Performance

Evaluating the unconstrained performance of the conditional Generative Adversarial Network (cGAN) was executed over a total training duration of 150 epochs. These 150 epochs yield a model ensemble whose global macro-properties are analyzed across 100 stochastically generated realizations. These 100 generated realizations exhibit a global facies volume distribution that appears practically identical to the baseline training suite, as illustrated in Figure 4.3.

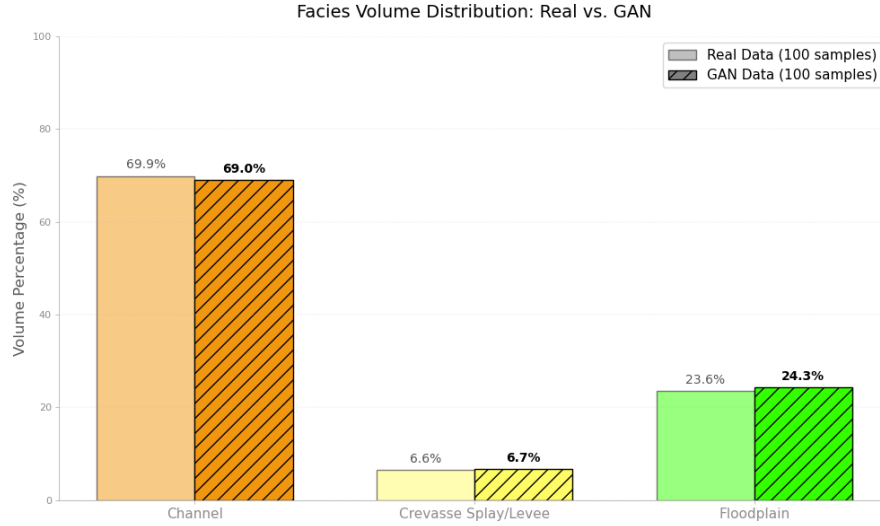


Figure 4.3: Comparative analysis of global facies volume percentages between the true training dataset and 100 unconstrained cGAN realizations, demonstrating macro-scale categorical matching.

While this identical volume distribution suggests successful macro-scale adaptation, a localized spatial evaluation reveals a structural failure to capture true sample diversity. This failure to capture sample diversity is visually apparent in the XY-plane normalized cell-wise entropy plots presented in Figure 4.4.

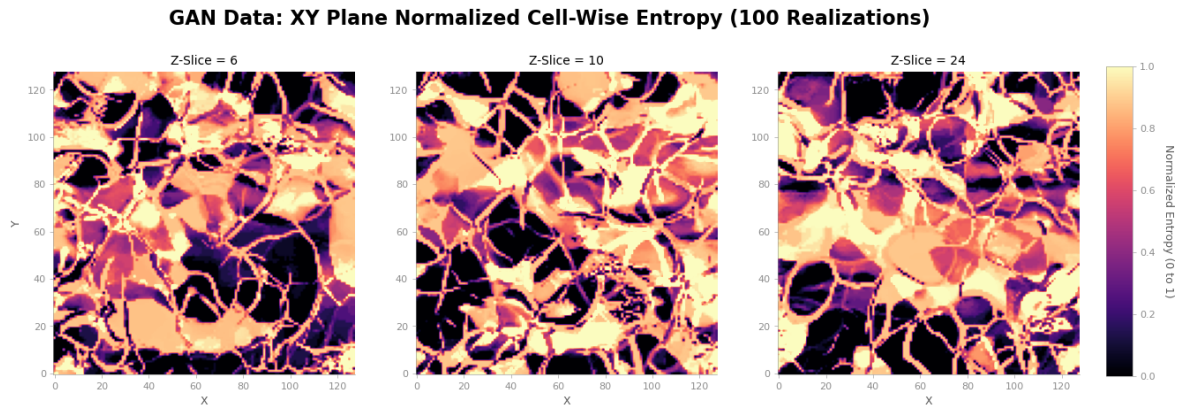


Figure 4.4: Normalized cell-wise spatial entropy across selected horizontal slices ($Z = 6, 10, \text{ and } 24$) calculated over 100 unconstrained cGAN realizations.

These 3D spatial entropy plots show heavily restricted uncertainty across the simulated grid slices, a trend further substantiated by the vertical axis statistics compiled in Table 4.1.

Table 4.1 highlights a severe mathematical discrepancy, where the GAN normalized entropy drops to approximately 0.55–0.59 compared to the near-maximum uncertainty values preserved in the true dataset. This low GAN entropy, combined with a persistent Jensen-Shannon Divergence (JSD) value

Slice Axis	Voxel Index	Real Normalized Entropy	GAN Normalized Entropy	JSD (Real vs. GAN)
Z	0	1.014740	0.551894	0.290421
Z	15	0.960197	0.590325	0.288134
Z	31	0.983518	0.587173	0.301107

Table 4.1: Spatial uncertainty and distribution divergence metrics along the vertical (Z) axis for the baseline training data and unconstrained cGAN realizations trained on Setting 1.

around 0.29, mathematically confirms that the network has entered a state of severe mode collapse. This severe mode collapse restricts the generator’s stochastic capability, causing the framework to repeat specific spatial configurations over and over instead of exploring the full architectural variety of the geological prior.

4.2.3. WGAN Unconstrained Performance

4.2.4. High-Fidelity 9-Class Generative Performance

4.3. Post-Inversion Conditioning Results

4.3.1. cGAN Latent Space Optimization and Pivotal Tuning

4.3.2. WGAN Latent Space Optimization and Pivotal Tuning

5 Discussion & Recommendations

5.1. Interpretations of key results

The implementation of the Fluvgan framework demonstrated highly stable training dynamics throughout the experimental phase. Once an optimal network architecture was established—a process that proved to be relatively straightforward—the generator and discriminator maintained a steady loss equilibrium. This stability is particularly notable given the high-dimensional complexity of 3D spatial data, confirming that the utilized residual block architecture effectively manages the transition from latent space to high-resolution facies models.

However, evaluating the unconstrained conditional GAN (cGAN) reveals a stark contrast between macro-scale replication and true spatial complexity. As demonstrated by the global volume statistics, the cGAN successfully mimics the overall facies distributions of the geological prior, producing structural proportions that appear nearly identical to the training data. Yet, spatial entropy metrics indicate that the model lacks a rigorous understanding of the true complexity inherent in the synthetic datasets. The severe drop in normalized spatial entropy mathematically confirms a state of mode collapse, wherein the network repeatedly generates similar spatial configurations rather than exploring the full architectural diversity of the fluvial-deltaic system.

Despite these generative limitations in unconstrained scenarios, the model excelled during the post-training conditioning phase. The two-step inversion process—combining latent space optimization with pivotal tuning—was highly successful in anchoring the generated realizations to hard well data. Notably, several optimized latent spaces achieved an average structural match of 99% against the empirical well data. This exceptional conditioning accuracy validates the practical feasibility of using GAN inversion to force mathematically bounded, stochastically generated reservoirs to honor localized, real-world subsurface observations.

5.2. Limitations of the current work

Severe Class Imbalance: The foundational FLUMY training datasets are characterized by extreme volumetric imbalance. Dominant channel structures overshadow intermediate facies (such as the Levee & Crevasse association), resulting in near-zero occurrence rates for specific minority classes. This statistical suppression inherently bottlenecks the inversion process, as the network struggles to accurately represent structures it rarely encountered during training.

Ensemble Mode Collapse: The cGAN's susceptibility to mode collapse fundamentally limits its utility for comprehensive uncertainty quantification. Because the model fails to capture the full stochastic diversity of the training data, the resulting ensemble underestimates the true geological uncertainty present between sparse data points.

Grid Resolution and Scaling Constraints: The fixed voxel resolution of 128x128x32 restricts the simulated domain to a localized 2560m x 2560m x 32m volume. While sufficient for demonstrating proof-of-concept well conditioning between two nearby wellbores, this grid size cannot encapsulate the entire Delft Sandstone reservoir, presenting computational challenges for full-field upscaling.

5.3. Recommendations for future work

Full-Field Upscaling and Dynamic Simulation: Research should expand beyond static geological evaluation by scaling the generated volumes to full-field dimensions using patch-based generation or multi-scale GANs. Furthermore, the 99% conditioned realizations should be subjected to dynamic fluid flow simulations to quantify how the ML-generated architectures impact actual geothermal heat extraction and fluid connectivity compared to traditional geostatistical models.

6 Conclusion

6.1. Answers to Research Questions

6.2. Final Remarks

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A FLUMY dataset generation

Value	Facies Key	Color	Description
0	Background		Undefined
1	Channel Lag		Coarse active channel fill
2	Point Bar		Meandering lateral accretion
3	Sand Plug		Fine-grained channel abandonment
4	Crevasse Splay I		Proximal high-energy overbank
5	Crevasse Channel		Feeder for splay deposits
6	Crevasse Splay II		Feeder for splay deposits
7	Levee		Bordering channel ridges
8	Overbank		Vegetated floodplain silt
9	Mud plug		Organic-rich anoxic sediments
10	Hemipelagic plug		Fine-grained marine deposits consisting of clay and silt-sized grains
11	Wetland/Paleosol		Organic-rich anoxic sediments
12	Draping		Fine-grained sediments conforming to topography
13	Pelagic		Deep-marine fine-grained suspension settling

Table A.1: Overview of the 9 facies classes used in the training datasets.

B GAN Training

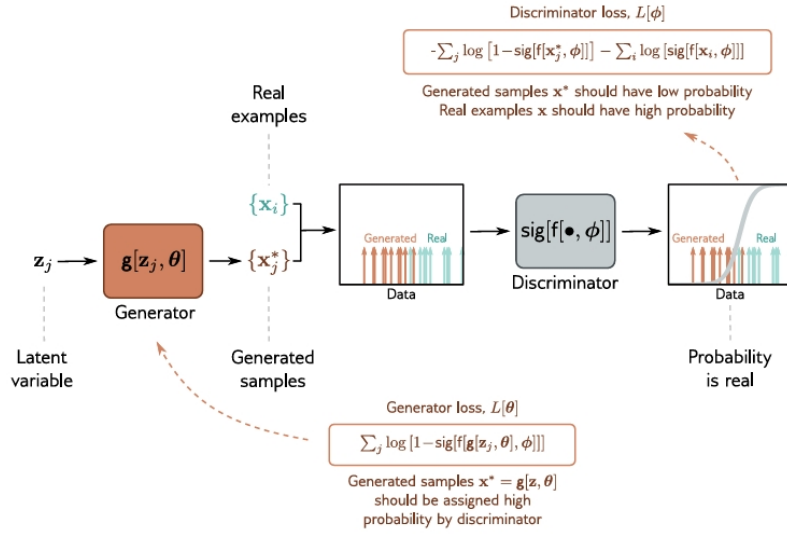


Figure B.1: Basic overview of a standard GAN architecture with a Generator, constructing fake samples from a random latent variable and a Discriminator deciding which of the two samples is the real sample (Prince, 2023).

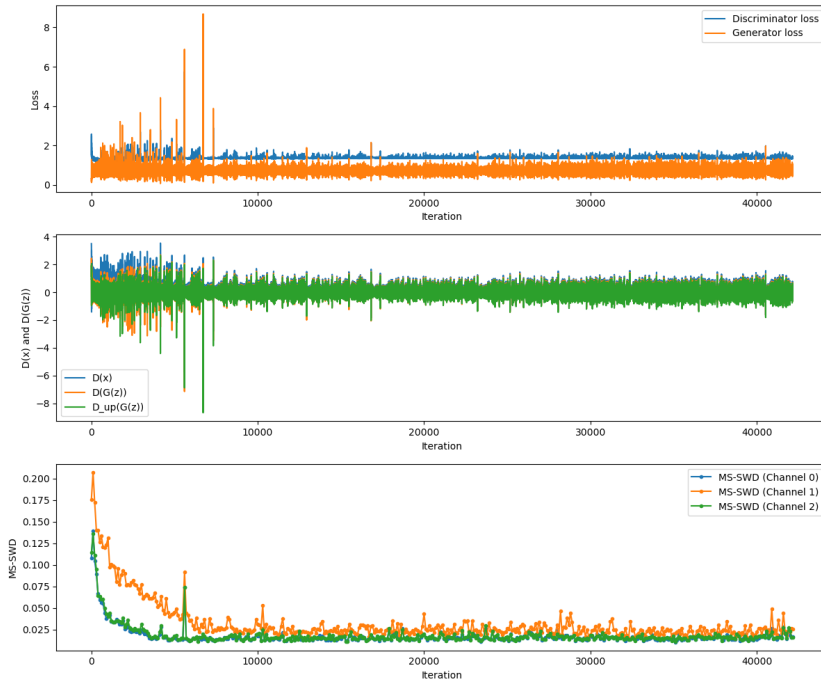


Figure B.2